

Periodic Table of Elements

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Z mass Symbol Name black: natural gray: man-made																	
11.009180 1 H Hydrogen 6.941 3 Li Lithium 9.0122 4 Be Beryllium 22.990 11 Na Sodium 24.305 12 Mg Magnesium 39.098 19 K Potassium 40.078 20 Ca Calcium 44.956 21 Sc Scandium 47.867 22 Ti Titanium 50.942 23 V Vanadium 51.996 24 Cr Chromium 54.938 25 Mn Manganese 55.845 26 Fe Iron 58.933 27 Co Cobalt 58.693 28 Ni Nickel 63.546 29 Cu Copper 65.39 30 Zn Zinc 69.723 31 Ga Gallium 72.64 32 Ge Germanium 74.922 33 As Arsenic 78.96 34 Se Selenium 79.904 35 Br Bromine 83.8 36 Kr Krypton 85.468 37 Rb Rubidium 87.62 38 Sr Strontium 88.906 39 Y Yttrium 91.224 40 Zr Zirconium 92.906 41 Nb Niobium 95.94 42 Mo Molybdenum 96 43 Tc Technetium 101.07 44 Ru Ruthenium 102.91 45 Rh Rhodium 106.42 46 Pd Palladium 107.87 47 Ag Silver 112.41 48 Cd Cadmium 114.82 49 In Indium 118.71 50 Sn Tin 121.76 51 Sb Antimony 127.6 52 Te Tellurium 126.9 53 I Iodine 131.29 54 Xe Xenon 132.91 55 Cs Caesium 137.33 56 Ba Barium 178.49 72 Hf Hafnium 180.95 73 Ta Tantalum 183.84 74 W Tungsten 186.21 75 Re Rhenium 190.23 76 Os Osmium 192.22 77 Ir Iridium 195.08 78 Pt Platinum 196.97 79 Au Gold 200.59 80 Hg Mercury 204.38 81 Tl Thallium 207.2 82 Pb Lead 208.98 83 Bi Bismuth 209 84 Po Polonium 210 85 At Astatine 222 86 Rn Radon 223 87 Fr Francium 226 88 Ra Radium 238.03 92 U Uranium 237 93 Np Neptunium 244 94 Pu Plutonium 243 95 Am Americium 247 96 Cm Curium 247 97 Bk Berkelium 251 98 Cf Californium 252 99 Es Einsteinium 257 100 Fm Fermium 258 101 Md Mendelevium 259 102 No Nobelium 262 103 Lr Lawrencium 138.91 57 La Lanthanum 140.12 58 Ce Cerium 140.91 59 Pr Praseodymium 144.24 60 Nd Neodymium 145 61 Pm Promethium 150.36 62 Sm Samarium 151.96 63 Eu Europium 157.25 64 Gd Gadolinium 158.93 65 Tb Terbium 162.50 66 Dy Dysprosium 164.93 67 Ho Holmium 167.26 68 Er Erbium 168.93 69 Tm Thulium 173.04 70 Yb Ytterbium 174.97 71 Lu Lutetium 227 89 Ac Actinium 232.04 90 Th Thorium 231.04 91 Pa Protactinium 238.03 92 U Uranium 237 93 Np Neptunium 244 94 Pu Plutonium 243 95 Am Americium 247 96 Cm Curium 247 97 Bk Berkelium 251 98 Cf Californium 252 99 Es Einsteinium 257 100 Fm Fermium 258 101 Md Mendelevium 259 102 No Nobelium 262 103 Lr Lawrencium																	

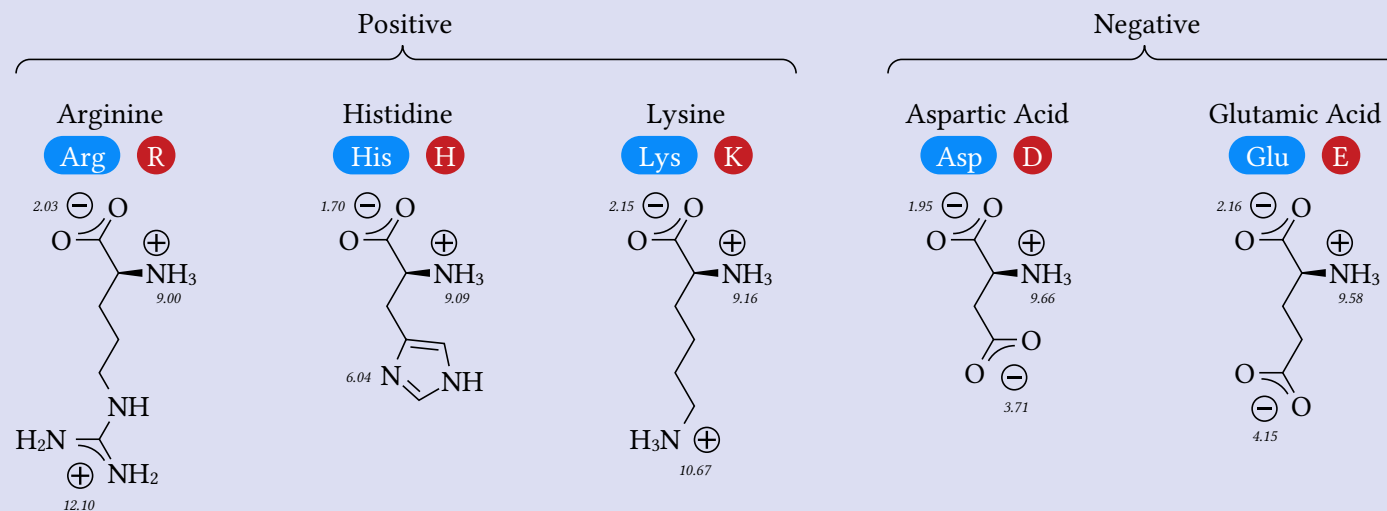
TWENTY-ONE PROTEINOGENIC α -AMINO ACIDS

Side chain charge
at physiological
pH 7.4

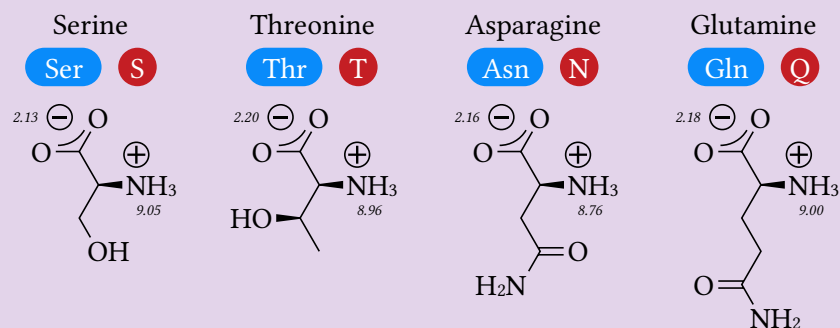
pK_a values shown
italicized

⊕ Positive
⊖ Negative

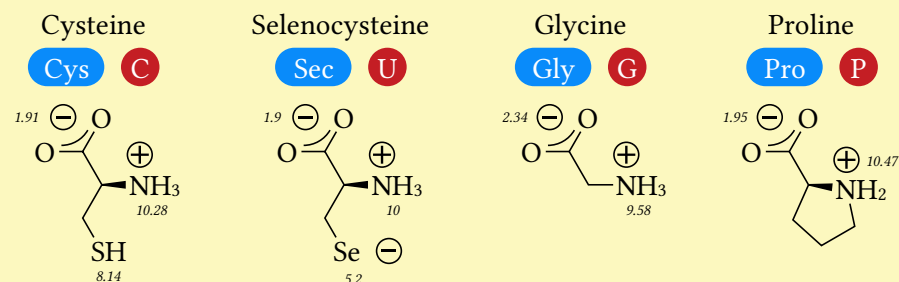
A. Amino Acids with Electrically Charged Side Chains



B. Amino Acids with Polar Uncharged Side Chains



C. Special Cases



D. Amino Acids with Hydrophobic Side Chains

